14.3. 作闭合回路：

$$U = \int_0^{\omega R} B\omega r \cdot dr = \frac{1}{2}\omega B \cdot \frac{1}{2}R^2 = \frac{1}{2}\omega R^2 B.$$

O点电势高.

14.5 a端电势高.

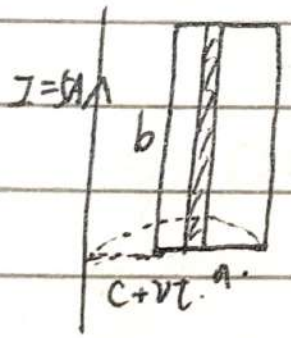
$$B = \frac{\mu_0 I}{2\pi r}.$$

$$U = \int_{0.1}^{1} B \cdot v \cdot dr = \int_{0.1}^{1} \frac{\mu_0 I}{2\pi r} v\, dr = \frac{4\pi\times10^{-7}\times2\times4}{2\times\pi} \ln 10$$

$$= 5.68\times10^{-6}\ V.$$

14.6 $U = \int_0^{L\sin\theta} B \cdot \omega r \sin\theta\, d(r\sin\theta) = \frac{1}{2}\omega L^2 \sin^2\theta \cdot B.$ 1 导体棒下端电势高.

14.7.



$$\Phi = \int B ds \cdot N$$

$$= \int_{c+vt}^{c+a+vt} \frac{\mu_0 I}{2\pi r} \times dr \times b \cdot N.$$

$$= \frac{\mu_0 I N}{2\pi} \cdot \ln\left|\frac{c+a+vt}{c+vt}\right| \cdot \times b.$$

$$U = \frac{d\Phi}{dt} = \frac{\mu_0 I N}{2\pi} \cdot \frac{c+vt}{c+a+vt} \cdot \frac{v(c+vt) - v(c+a+vt)}{(c+vt)^2} \times b$$

$$t=0,\ U = \frac{\mu_0 I N}{2\pi} \times \frac{cb}{c+a} \cdot \frac{vc - v(c+a)}{c^2} = \frac{\mu_0 I N}{2\pi} \cdot \frac{-vab}{(c+a)c}$$

$$= \frac{4\pi\times10^{-7}\times5\times1000}{2\pi} \times \frac{-0.03\times0.02\times0.04}{0.07\times0.03} = 6.86\times10^{-6}\ V.$$

或 $U = U_1 - U_2 = NB_1 b v - NB_2 b v$，方向为顺时针方向

14.9. $\Phi = \int B \cdot dS = \int_0^{0.2} 4t^2 y \cdot 0.2 \, dy = 0.8t^2 \cdot \frac{1}{2} y^2 \Big|_0^{0.2} = 1.6 \times 10^{-2} \times t^2$

$U = \frac{d\Phi}{dt} = 2 \times 1.6 \times 10^{-2} \times t = 3.2 \times 10^{-2} \times 0.25 = 8 \times 10^{-3} V$

方向为顺时针方向.

14.11 $dU = -\frac{dB}{dt} \cdot \pi r^2 = -k \cdot \pi r^2$

$R = \rho \cdot \frac{2\pi r}{b\,dr}$

$dI = \frac{dU}{R} = \frac{-k\pi r^2}{\rho \cdot \frac{2\pi r}{b\,dr}} = \frac{-k\cancel{\pi} r^{\cancel{2}}}{\rho \cdot 2\cancel{\pi r}} \cdot b \cdot dr = \frac{-bkr\,dr}{2\rho}$

$I = \int_0^a \frac{-bkr\,dr}{2\rho} = -\frac{bk}{2\rho} \cdot \frac{1}{2} a^2 = \frac{-a^2 bk}{4\rho}$

14.15 (1). $U_{ad} = 0$, $U_{bc} = 0$

$U_{dc} = \frac{dB}{dt} \cdot \frac{\sqrt{3}}{4} \times \left(\frac{R}{2}\right)^2 = \frac{\sqrt{3}}{16} R^2 \cdot \frac{dB}{dt}$，c点电势高

$U_{ab} = \frac{dB}{dt} \cdot \frac{\sqrt{3}}{4} \times R^2 = \frac{\sqrt{3}}{4} R^2 \cdot \frac{dB}{dt}$，b点电势高

(2). $U = U_{ab} - U_{dc} = \frac{3}{16}\sqrt{3} R^2 \frac{dB}{dt}$.

14.17 (1) $B=\mu_0 nI$;

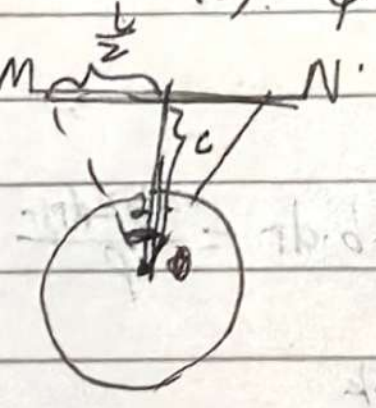
$$I=\frac{\omega q}{2\pi}=\frac{\beta t}{2\pi}\cdot\lambda L.$$

$$B=\frac{\mu_0\cdot\beta t\lambda}{2\pi}.$$

则 $E\cdot 2\pi b=-\frac{d\phi}{dt}=-\frac{\mu_0\beta\lambda}{2\pi}\cdot\pi b^2.$

得 $E=\frac{-\mu_0\beta\lambda b}{4\pi}$

(2). $\phi=\frac{2\alpha}{2\pi}\cdot B\cdot\pi a^2=\frac{\mu_0\beta t\lambda}{2\pi}\cdot\frac{2\alpha}{2}\cdot a^2=\frac{\mu_0\beta t\lambda a^2\alpha}{2\pi}.$



$$U=-\frac{d\phi}{dt}=-\frac{\mu_0\beta\lambda a^2\alpha}{2\pi}.$$

$\tan\alpha=\frac{\frac{L}{2}}{c}=\frac{L}{2c}$，则 $\alpha=\arctan\frac{L}{2c}.$

故 $U=-\frac{\mu_0\beta\lambda a^2\alpha}{2\pi}\cdot\arctan\frac{L}{2c}$，N电势高.